

Mathematical Derivation: Capillary Sorption with Bubble Release Phenomena

The Core Variable Transformation (Length to Mass)

Standard Washburn equations use penetration depth (L). Since the experimental data (tensiometer) records mass (m), we must first establish the conversion.

The mass of liquid absorbed is a function of the tube's cross-sectional area (A), powder porosity (φ), liquid density (ρ), and penetration depth (L):

$$m(t) = L(t) \cdot A \cdot \varphi \cdot \rho$$

Rearranging for $L(t)$ and taking the derivative with respect to time gives us our substitution keys:

$$L = \frac{m}{A\varphi\rho}$$

$$\frac{dL}{dt} = \frac{1}{A\varphi\rho} \frac{dm}{dt}$$

Derivation: The Modified Setup (Hollow Tube)

Condition: Air escapes freely. Atmospheric pressure is maintained inside the powder bed. The opposing back-pressure of air (P_{back}) is strictly 0.

We start with the standard Washburn differential force balance:

Viscous Drag = Capillary Pressure

$$\frac{8\eta L}{r^2} \frac{dL}{dt} = \frac{2\gamma \cos\theta}{r}$$

Substitute the mass variables derived in Section 1:

$$\frac{8\eta}{r^2} \left(\frac{m}{A\phi\rho} \right) \left(\frac{1}{A\phi\rho} \frac{dm}{dt} \right) = \frac{2\gamma \cos\theta}{r}$$

Simplify and isolate the differential terms:

$$m \frac{dm}{dt} = \frac{r(\rho A\phi)^2 \gamma \cos\theta}{4\eta}$$

(This is the modified ODE for the hollow tube in mass form).

To find the **Analytical Solution**, integrate both sides with respect to t :

$$\int m \, dm = \int \frac{r(\rho A\phi)^2 \gamma \cos\theta}{4\eta} \, dt$$

$$\frac{m^2}{2} = \left[\frac{r(\rho A\phi)^2 \gamma \cos\theta}{4\eta} \right] t$$

Multiply by 2 to get the final **Analytical Equation**:

$$m^2(t) = \left[\frac{r(\rho A \phi)^2 \gamma \cos \theta}{2\eta} \right] t$$

By grouping the geometric constants of the powder bed into a single structural constant

$c = \frac{r(\rho A \phi)^2}{2}$, we arrive at the final closed-form equation:

$$m^2(t) = \left(\frac{c \cdot \gamma \cos \theta}{\eta} \right) t$$

(This proves mathematically that a perfectly unblocked capillary system will yield a linear m^2 vs t curve).

Derivation: The Original Setup (1mm Perforated Plate)

Condition: The 1mm holes form a liquid seal. Trapped air volume (V_{air}) decreases as water enters, creating a time-dependent back-pressure (P_{back}) opposing the capillary action.

Step A: Define the Back-Pressure

Initial air volume is V_0 . As mass m enters, the available air volume shrinks:

$$V_{air}(m) = V_0 - \frac{m}{\rho}$$

By Boyle's Law ($P_1 V_1 = P_2 V_2$), the absolute pressure of the trapped air becomes:

$$P_{air}(m) = P_0 \left(\frac{V_0}{V_0 - m/\rho} \right)$$

The effective back-pressure opposing the flow is the difference between trapped air and external atmospheric pressure:

$$P_{back}(m) = P_{air}(m) - P_0 = P_0 \left(\frac{m/\rho}{V_0 - m/\rho} \right)$$

Step B: The Modified ODE (Mass Form)

We return to the Washburn force balance, but now subtract the back-pressure:

$$\text{Viscous Drag} = \text{Capillary Pressure} - \text{Air Back} - \text{Pressure}$$

$$\frac{8\eta L}{r^2} \frac{dL}{dt} = P_c - P_{back}(m)$$

Substitute the mass variables (L and dL/dt):

$$\frac{8\eta}{r^2 (A\phi\rho)^2} \left(m \frac{dm}{dt} \right) = \frac{2\gamma\cos\theta}{r} - P_0 \left(\frac{m/\rho}{V_0 - m/\rho} \right)$$

This is the **Governing ODE for the Failed Setup**:

$$m \frac{dm}{dt} = \frac{r^2 (A\phi\rho)^2}{8\eta} \left[\frac{2\gamma\cos\theta}{r} - P_0 \left(\frac{m}{\rho V_0 - m} \right) \right]$$

Step C: The Analytical Solution (Implicit Form)

Unlike the hollow tube, this ODE is non-linear but separable. It can be analytically integrated to find an implicit equation for time t as a function of mass m .

Let the grouped constants be $K = \frac{r^2(A\phi\rho)^2}{8\eta}$ and Capillary Pressure be P_c .

Separating variables yields the integral:

$$t(m) = \frac{1}{K} \int \frac{m(\rho V_0 - m)}{P_c \rho V_0 - (P_c + P_0)m} dm$$

This integral resolves into a closed-form algebraic equation using polynomial division:

$$t(m) = C_1 m + C_2 m^2 + C_3 \ln \left(1 - \frac{P_c + P_0}{P_c \rho V_0} m \right)$$

(Where C_1, C_2, C_3 are algebraic combinations of K, P_c, P_0, ρ, V_0 . This logarithmic term mathematically dictates the exact curve where sorption flattens out and halts).

Step D: The Bubble Burst Condition

The analytical curve above describes the sorption *only until* the pressure is high enough to force a bubble out of the 1mm hole. Mathematically, the bubble escapes when Laplace pressure is exceeded:

$$P_{air}(m) \geq P_0 + \frac{2\gamma}{r_{hole}}$$

When this threshold is reached, V_{air} resets to V_0 , causing the discontinuous jumps seen in the tensiometer data.

Computational Validation of Capillary Sorption Dynamics

Attached below are the computational model and resultant plots developed using Python. This script utilizes the closed-form analytical mass equations derived from the modified Washburn force balance.

```
import numpy as np
import matplotlib.pyplot as plt

# --- Physical Parameters ---
gamma = 0.0728      # Surface tension of water (N/m)
eta = 0.001         # Viscosity (Pa.s)
rho = 1000.0        # Density (kg/m3)
P0 = 101325.0       # Atm pressure (Pa)

# --- Calibrated Powder Constants ---
c_powder = 7.51e-17
r_pore = 30e-9      # Corresponding effective pore radius (~30 nm)
Pc = (2 * gamma) / r_pore # Capillary pressure

# Ideal Washburn Slope in SI (kg^2/s) based on your real powder
ideal_slope_si = (c_powder * rho**2 * gamma) / eta

# --- Integration Parameters ---
V0 = 5e-7 # Estimated trapped air volume (0.5 mL)

# EXTENDED TIME: Now simulating 300 seconds (5 minutes)
t_max = 300
times = np.linspace(0, t_max, 2000)
dt = times[1] - times[0]

# 1. Ideal Hollow Tube (Linear m^2)
m2_ideal_si = ideal_slope_si * times
m2_ideal_g2 = m2_ideal_si * 1e6 # Convert to g^2

# 2. Blocked 1mm Plate Setup (Logarithmic Halt)
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m2_blocked_si = [0.0]

for t in times[1:]:
    m2_current = m2_blocked_si[-1]

    # Prevent negative square roots from numerical noise
    if m2_current < 0: m2_current = 0
    m_current = np.sqrt(m2_current) # Current mass in kg

    # Calculate Back-pressure from Boyle's Law
    V_water = m_current / rho
    if V_water >= V0: V_water = V0 - 1e-12 # Prevent division by zero

    P_back = P0 * (V_water / (V0 - V_water))

    # Modified Washburn ODE integrated for m^2 directly:
    # d(m^2)/dt = Ideal_Slope * (1 - P_back / Pc)
    pressure_ratio = P_back / Pc

    if pressure_ratio > 1.0:
        pressure_ratio = 1.0 # Sorption halts completely when back-pressure
        overpowers capillary pressure

    dm2_dt = ideal_slope_si * (1.0 - pressure_ratio)

    m2_next = m2_current + (dm2_dt * dt)
    m2_blocked_si.append(m2_next)

m2_blocked_g2 = np.array(m2_blocked_si) * 1e6

# --- Plotting ---
plt.figure(figsize=(10, 6))
plt.plot(times, m2_ideal_g2, 'g--', label='Analytical Hollow Tube (Linear $m^2$)', linewidth=2)
plt.plot(times, m2_blocked_g2, 'r-', label='ODE 1mm Plate (Logarithmic Halt)', linewidth=2)
plt.title("Mathematical Derivation Proof: Mass Squared vs Time (Extended)")
plt.xlabel("Time [s]")
plt.ylabel("Mass Squared [g²]")
plt.legend()
plt.grid(True)
plt.show()

```

